Computer Networks
Fall 2024/25
SOLUTION
Exercise-1

Problem 1 (15 points).

1. Answer**: 6.5 × 1000 × 8 / 1000000 = 0.052sec = 52msec**

2. Answer: D = $(2.5\times10^8) \times 80/(100\times10^3)$ = 200,000 meters.

3. Answer: Yes.

Problem 2 (20 points).

Answer:

1. Verify the students have been using 'tracert' properly. ANY measurements are acceptable as long as they show the student has understood the usage of tracert.
2. They are all IP addresses of **Free & Public DNS Servers.**

Problem 3 (20 points).

Answer: Try to focus on how much time it takes for the last packet.

Propagation Delay: $10^9 / 2*10^8$ = **5** seconds**.**

Transmission Delay from A: 200/10 =**20** seconds

The last packet leaves R1 after: 100-18 = **82** seconds (since it arrived at R1).

Store-and-Forward Delay (for the 11 routers/ transmission rate 5): 11*20/5 = **44** seconds.

Total Delay = 20+82+44+5 = **151 seconds**.

Problem 4 (20 points).

Answer: The students should attach a proper HTML page & a screen capture.

Problem 5 (25 points). Enclosed, please find the documents "Wireshark Lab: Getting Started." Answer these **4** questions. Check answers.